

Dukaan

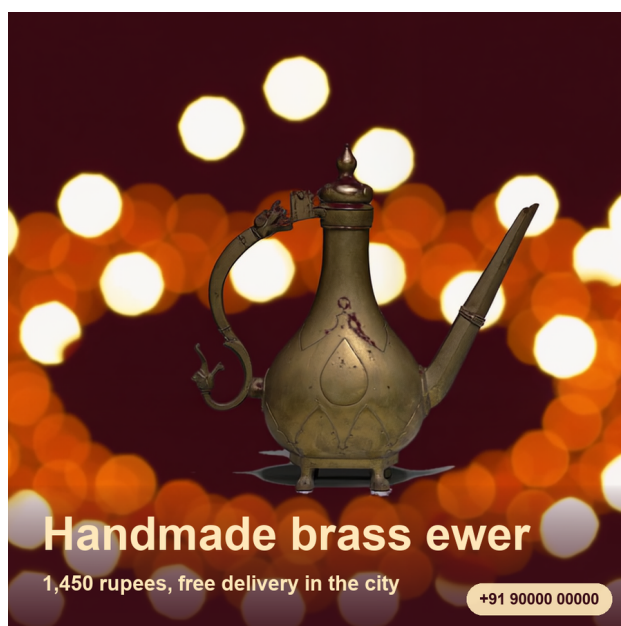
One product photo in, a shop's worth of creatives out. Generated on AMD Radeon.

Track 1, Development of Multimodal Content Creation Tools himanshu748 | Radeon PRO gfx1100, 48 GB | ROCm 7.2.4 | LTX-2.3 22B

The problem

A shopkeeper photographs stock in ten seconds. Turning that into something postable takes an afternoon in a design tool, per product, every time the catalogue turns over. Tools that close the gap usually generate the product too, which is the wrong trade: a seller cannot post a picture of a bangle that is not the bangle they will ship.

What Dukaan does



One photo becomes a square for the feed, a 9:16 for stories, a wide banner for a shop header, and a short clip with generated ambient audio. The product is carried through untouched. The price and phone number are composited with real type, never drawn by a model.

The GPU runs once per pack, not once per format. Three formats are three different moments of the same generated clip.

The finding: the instance cannot run its own template

LTX-2.3 checkpoint	43 GB
Gemma 3 12B text encoder	23 GB
Total to load	66 GB
Container cap	55 GB

ComfyUI holds every model a graph touches for the life of the process. Loading both trips the cap and **the platform restarts the container mid-prompt**: JupyterLab returns with zero kernels, the port stops answering, and it presents as a network fault.

The fix: two processes that never overlap

phase	peak container RAM	wall
encode, text encoder only	35.4 GB	33 s
sample, checkpoint only	51.2 GB	55 s
stock template, one process	trips 55 GB, restarts	n/a

Peak becomes `max(43, 23)`, not `43 + 23`.

Plus: bf16 conditioning (it is read back while the checkpoint is resident), VRAM eviction before the VAE decode (the decode died with 2.13 GB free out of 47.98), and `torch.inference_mode()` around the phase (the LTX VAE updates the sampler's output in place).

Measured

product	GPU	wall	output
brass ewer	65.3 s	2 m 20 s	3 stills, 49 frames, 1.96 s stereo
silver bangle	72.6 s	2 m 19 s	3 stills, 49 frames, 1.96 s stereo
gilt bangles	62.0 s	batched	3 stills, 49 frames, 1.96 s stereo

Transport was a bottleneck too: 49 frames as 49 base64 requests took 3 m 34 s and reset the tunnel twice. One tar brought it to 2 m 19 s.

Three CPU decisions that keep GPU credits for the GPU

The cutout fits the background, it does not sample it. A quadratic surface per channel on the border ring, refit once with the worst residuals dropped. A plane left an elliptical pool of backdrop exactly where the product sits.

Stills are picked for spread, then sharpness. Contiguous windows, highest Laplacian variance in each. Frame 0 is never eligible: it is the plate the model was handed.

Reshaping mirrors, it does not crop or stretch. Cover-cropping a square to 9:16 cuts the product; clamping the edge row drew the bokeh into horizontal streaks.



26 tests, no GPU required. Without an instance configured, the whole pipeline runs on CPU and writes real files, so the tool can be inspected before any spend. Apache-2.0. Demo photographs are CC0.